

Academic Performance Task, Expectations and Rubric

The Task

This lesson looks at the Ottawa River Watershed in particular, using MFworks. It deals with building masks, created slope and shaded relief maps from DEM (Digital Elevation Model), composite overlays and data analysis. Students begin by examining various land use patterns, as they exist in the Ottawa River Watershed. Land use patterns include agriculture, urbanization, soils, and vegetation patterns as they relate to water quality. Students will research how different land use patterns affects water quality and write a report outlining various procedures, which could be imposed to improve water quality. The report should conclude by suggesting ways to prepare for the future, to insure that high water quality is available for future generations. Students are instructed to include supporting visuals, such as maps, graphs, charts, diagrams, or pictures, in the report.

Expectations

This task gives students the opportunity to demonstrate achievement of the following selected expectations from three strands: Geographic Foundations: Space and Systems; Understanding and Managing Change; and Methods of Geographic Inquiry.

Students will:

1. analyse factors that affect natural and human systems in Canada;
2. demonstrate an understanding of the challenges associated with achieving resource sustainability, and explain the implications of meeting or not meeting those challenges for future resource use in Canada;
3. explain the role of government in managing resources and protecting the environment.
4. synthesize information on changes in the geography of Canada, such as changes in land use and urban patterns, as well as resource depletion, in order to plan for the future;
5. identify and analyse patterns of spatial organization, including land use, population distribution, and ecozones;
6. create and use effectively photographs, charts, graphs, models, and diagrams
7. demonstrate an understanding of the methods used to collect, organize, manipulate, and interpret geographic data.

Prior Knowledge and Skills

To complete this task, students are expected to be able to demonstrate the following:

- know ways in which rivers are important to our lives and our culture;
- know about runoff, drainage basins, drainage divide, river flow, etc.;
- know about major Canadian water bodies;
- know about the hydrologic cycle;
- have an idea of water as a resource, Canada's supply of fresh water and the demands for water resources;
- have an idea of how rivers and water basins get contaminated;
- ability to read, analyse, and interpret satellite imagery and graphs;
- ability to use a variety of tools and technologies for research;
- select and use appropriate methods for displaying geographic data;
- use geographic data to support conclusions and opinions.

Task Rubric – Ottawa River Watershed, potential sources of pollution

Expectations	Criteria	Level 1 (50 – 59)	Level 2 (60 –69)	Level 3 (70 – 79)	Level 4 (80-100)
<u>Knowledge / understanding</u>					
1, 2	- Analyse factors that affect natural and human systems in Canada; - Demonstrate an understanding of the challenges associated with achieving resource sustainability, and explain the implications of meeting or not meeting those challenges for future resource use in Canada	- Shows little analyses of factors that affect natural and human systems - Demonstrates little understanding of the challenges associated with achieving resource sustainability	- Shows some analyses of factors that affect natural and human systems - Demonstrates some understanding of the challenges associated with achieving resource sustainability	- Shows considerable analyses of factors that affect natural and human systems - Demonstrates considerable understanding of the challenges associated with achieving resource sustainability	- Shows a high degree of analyses of factors that affect natural and human systems - Demonstrates a high degree of understanding of the challenges associated with achieving resource sustainability
<u>Thinking / Inquiry</u>					
4, 7	- Use a variety of relevant supporting facts and details - Explain the significance of the information selected	- Uses a limited variety of relevant supporting facts and details - Explains the significance to a limited degree	- Uses some variety of relevant supporting facts and details - Explains the significance to some degree	- Uses a considerable variety of relevant supporting facts and details - Explains the significance to a considerable degree	- Uses a wide variety of relevant supporting facts and details - Explains the significance to a high degree
<u>Communication</u>					
6	Use visuals to illustrate the importance of report	Uses visuals to illustrate the importance to a limited degree	Uses visuals to illustrate the importance to some degree	Uses visuals to illustrate the importance to a considerable degree	Uses visuals to illustrate the importance to a high degree
<u>Application</u>					
3	Explain the role of government in managing resources and protecting the environment.	Demonstrates little understanding	Demonstrates some understanding	Demonstrates considerable understanding	Demonstrates a high degree of understanding